

Where each factor comes from

Running Gram–Schmidt on the columns of A produces both factors at once. Q is the output and R is the bookkeeping — the coefficients that were subtracted along the way, kept rather than discarded.

TYPE	CONDITION	ENTRIES	PROPERTIES	WHAT IT RECORDS
THE OUTPUT				
	Q — orthonormal columns	$m \times n$	$\mathbf{q}_j = \mathbf{u}_j / \ \mathbf{u}_j\ $ $Q^T Q = I$	an orthonormal basis for $\text{Col}(A)$
THE BOOKKEEPING				
	R — upper triangular	$n \times n$	$r_{ij} = \mathbf{q}_i \cdot \mathbf{v}_j$ diagonal $r_{ii} = \ \mathbf{u}_i\ > 0$	the coefficients discarded during orthogonalization

This is why **QR** is not a separate algorithm from Gram–Schmidt so much as its record. Every projection coefficient computed during orthogonalization is an entry of R , and R is triangular precisely because $\mathbf{v}_j$ is only ever projected onto vectors that came before it.